Introduction to Mathematical Programming

1 Introductions

- Who am I?
- Student introductions: What year and what field? Research interests?
- What do you hope to get out of this course?

2 Course Orientation & Motivation

- Many economic questions ask an agent to make the best feasible choice: a farmer chooses a crop mix, a firm chooses output, a policymaker chooses an intervention, or a consumer chooses a bundle.
  - **Example: a farmer's crop-allocation decision**
    * The farmer chooses acres of wheat (x_w) and corn (x_c).
    * The objective is to maximize profit.
    * Available land and water limit the choices the farmer can make.

$$\begin{aligned} \max_{x_w, x_c} \quad & \pi_w x_w + \pi_c x_c \\ \text{s.t.} \quad & x_w + x_c \leq L && \text{(land)} \\ & a_w x_w + a_c x_c \leq W && \text{(water)} \\ & x_w, x_c \geq 0. && \text{(nonnegativity)} \end{aligned}$$

- This example illustrates the building blocks we will use throughout the course:
  - **Decision variables** are the choices under the decision maker's control (x_w, x_c).
  - The **objective function** states what the decision maker values (here, profit).
  - **Constraints** describe limits imposed by resources, technology, institutions, or policy.
  - The **feasible set** is the collection of choices that satisfy every constraint.
  - An **optimal solution** is the feasible choice that produces the best objective value.
- Some models have useful analytical solutions; many realistic models do not. We will use both analytical reasoning and numerical methods to formulate, solve, diagnose, and interpret optimization models.

3 What is a Mathematical Program?

- **General Form**:

$$\begin{aligned} \max_{x} \quad & f(x) \\ \text{s.t.} \quad & g_j(x) \leq 0, \quad j = 1, \dots, m, \\ & h_k(x) = 0, \quad k = 1, \dots, p, \\ & x \in X. \end{aligned}$$

- Components:
  - x is a vector of **decision variables** (for example, acres planted or quantities produced).
  - $f(x)$ is the **objective function** (for example, profit, cost, or utility).
  - $g_j(x) \leq 0$ are **inequality constraints**, such as resource limits or policy requirements.
  - $h_k(x) = 0$ are **equality constraints**, such as accounting identities or physical balance conditions.
  - X records simple restrictions on the decision variables, such as nonnegativity, bounds, or integrality.
- The farmer's crop-allocation model is one instance of this general form.

4 Where This Course Is Going

- By the end of the course, you should be able to:
  1. Formulate economic decision problems as numerical optimization models.
  2. Solve models in R and assess whether the solution is credible.
  3. Interpret solutions, constraints, and comparative statics in economic terms.
  4. Use optimization models to investigate applied research questions.
- We will begin with linear models, then study nonlinear and dynamic optimization. The same modeling questions will recur: What is the decision? What is the objective? What constrains it? What do the solution and its sensitivity mean?

5 Uses of Mathematical Programming (McCarl §1.3)

What does McCarl say are the uses of math programming?

1. **Problem Insight Construction**
   - State problem carefully and really understand the moving parts
   - Clarify objectives, constraints, and trade-offs.
   - Example: Writing down a water allocation model forces us to quantify resource limits. It also forces one to write down the functional relationships between variables, which may be a simplification or approximation of a physical or natural phenomenon: hydrology, population growth functions, atmospheric models.
2. **Numerical Applications**
   - **Prescription of Solutions**
     - Goal: Recommend an **optimal plan** given data, objectives, and constraints.
     - Example: Farm crop mix — how many acres of wheat, corn, soy to maximize profit?
   - **Prediction of Consequences**
     - Goal: Forecast **outcomes under scenarios**.
     - Example: If fertilizer availability drops 20%, what happens to optimal output and profit?
   - **Demonstration of Sensitivity**
     - Goal: Show how solutions **shift with parameters**.
     - Example: How does optimal irrigation use change as water prices rise?
3. **Algorithm Development**
   - We will not derive most solution algorithms from scratch, but we will learn enough about them to choose tools appropriately, recognize warnings, and assess solver output.

6 Preview: How Optimization Methods are Used in Research

- **Agricultural & Resource Economics**
  - Crop mix and land allocation models under climate or policy changes.
  - Water allocation across sectors or regions.
  - Forestry and rangeland management under biological growth constraints.
- **Environmental Economics**
  - Policy design problems: cap-and-trade, carbon taxes, renewable energy portfolios.
  - Nonpoint pollution control, groundwater management.
  - Climate-economy integrated assessment models.
- **Industrial Organization & Production**
  - Firm cost minimization and production planning.
  - Transportation and logistics networks.
  - Supply chain optimization.
- **Public Policy & Development**
  - Food aid allocation and trade policy.
  - Health policy interventions under budget constraints.
  - Energy access and electrification planning.

Takeaway: *Mathematical programming is not just a classroom exercise; it's a core toolkit for applied economic research.* The methods we learn will connect directly to how we ask and answer research questions in environmental, agricultural, and resource economics.